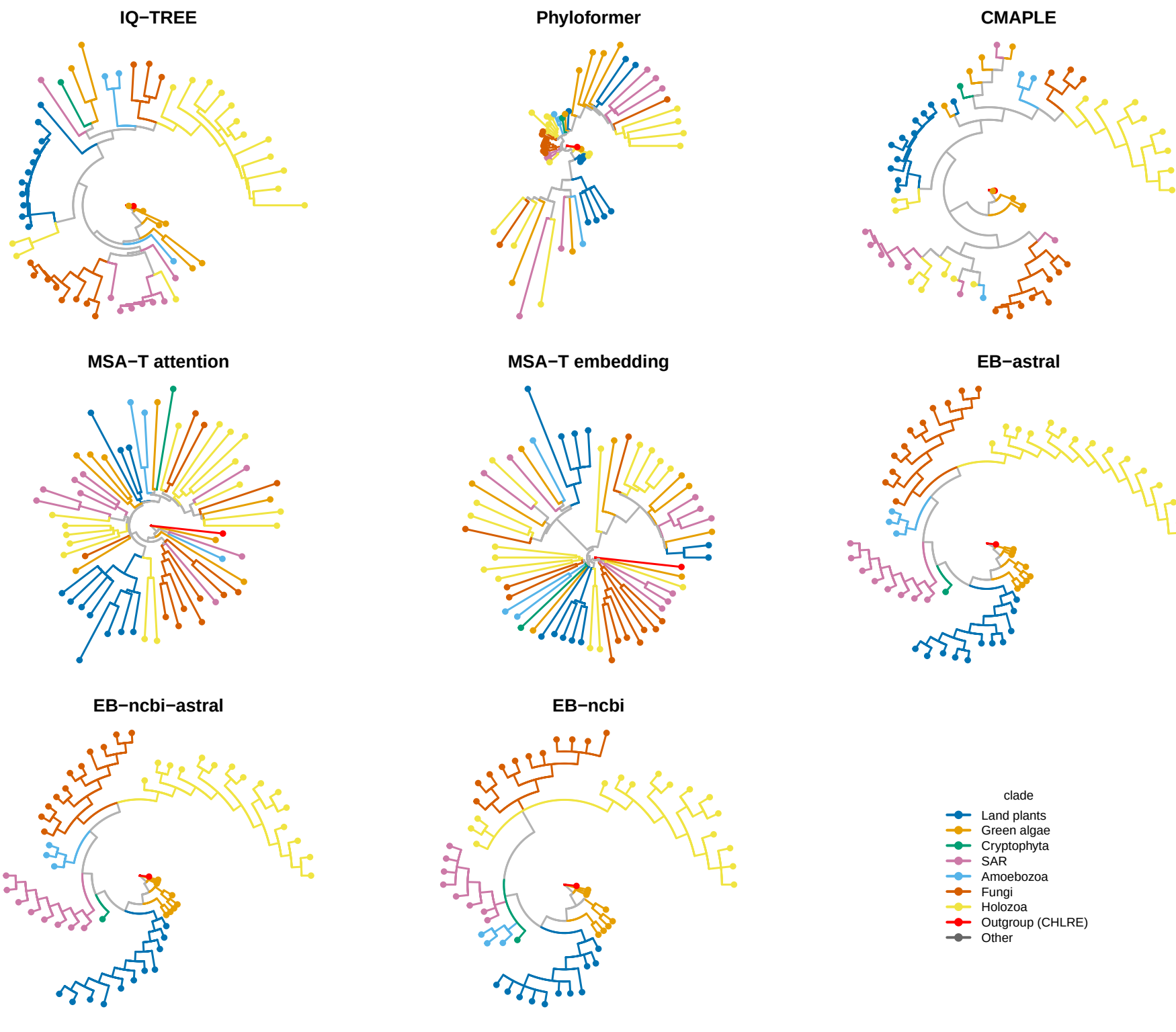
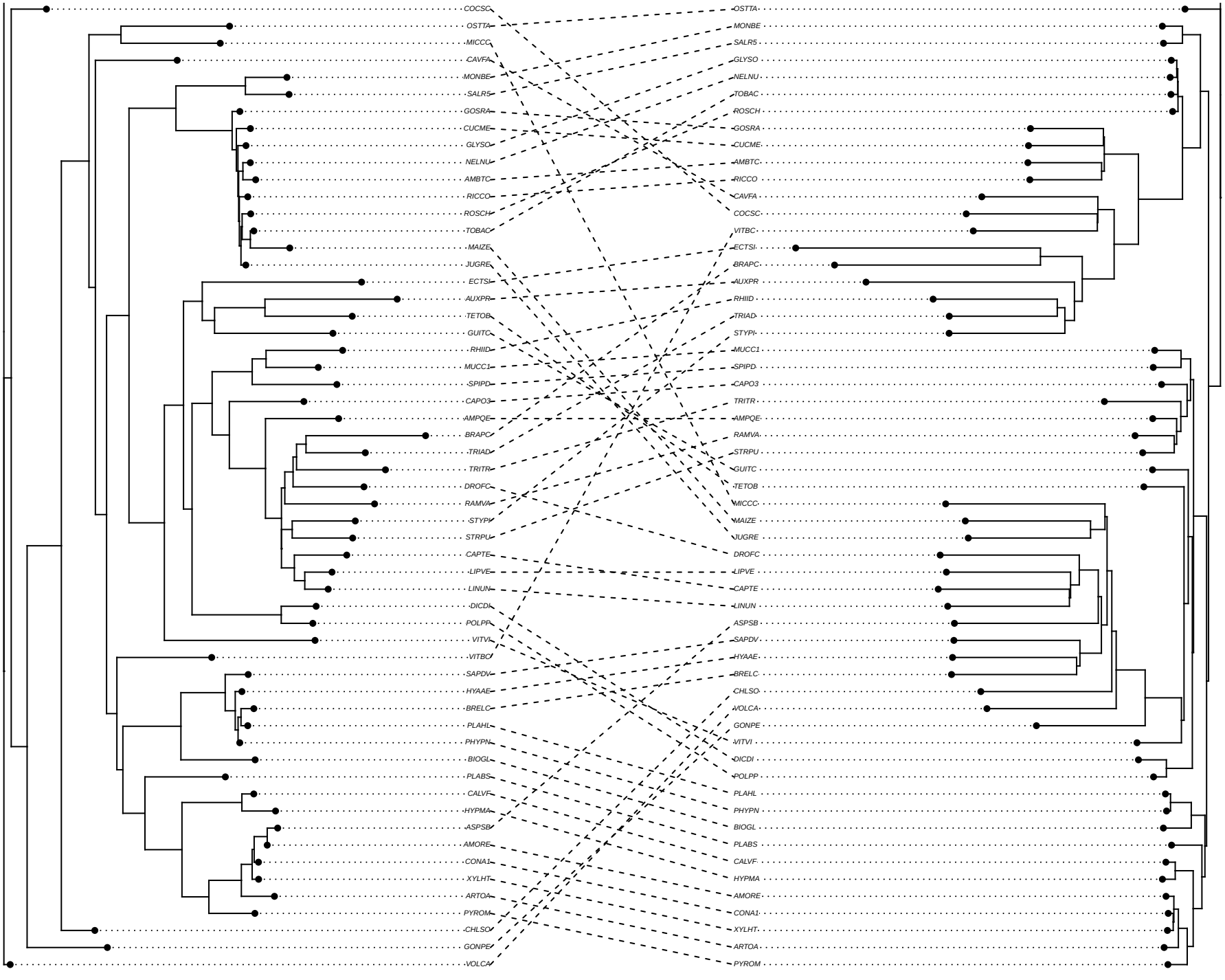
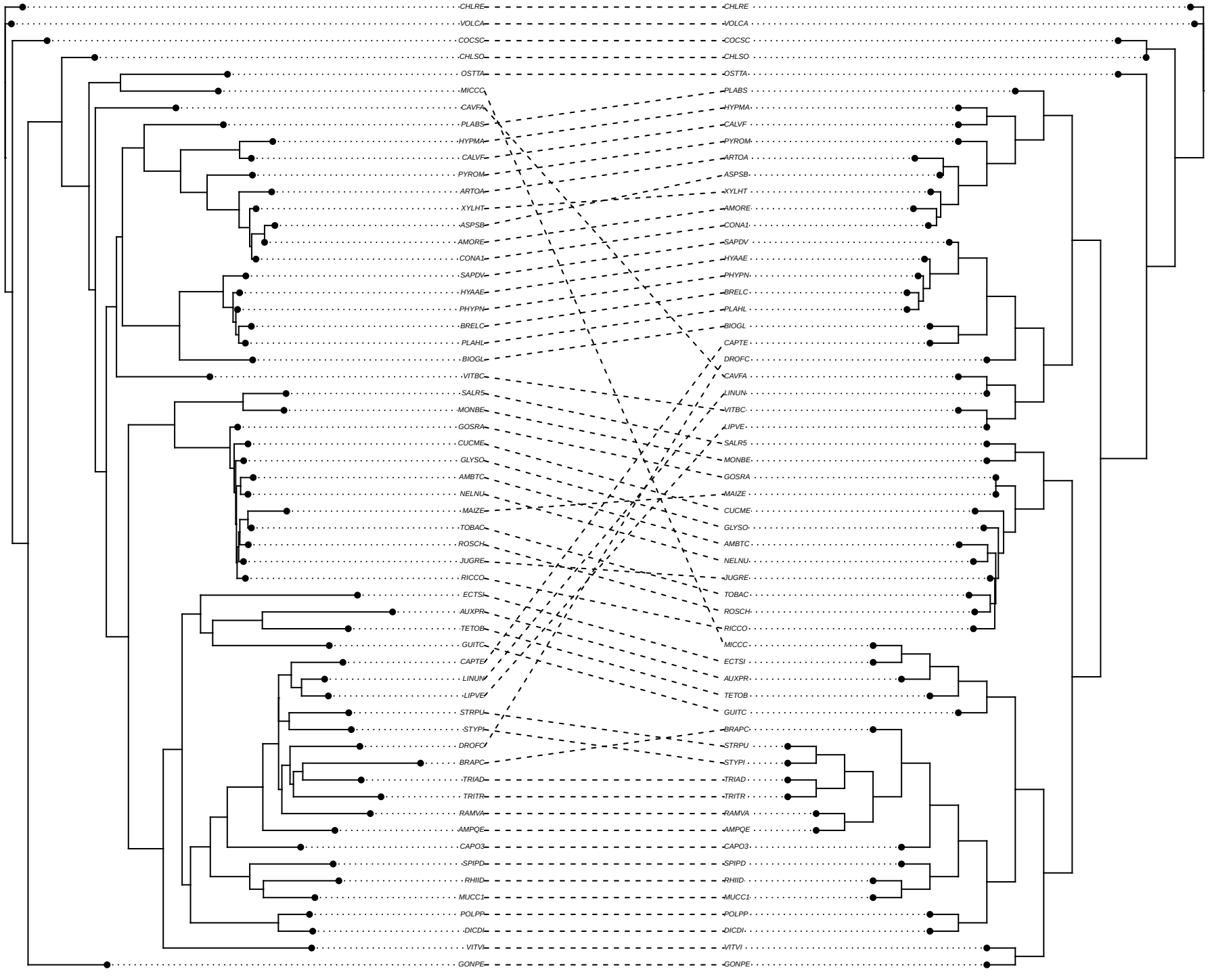
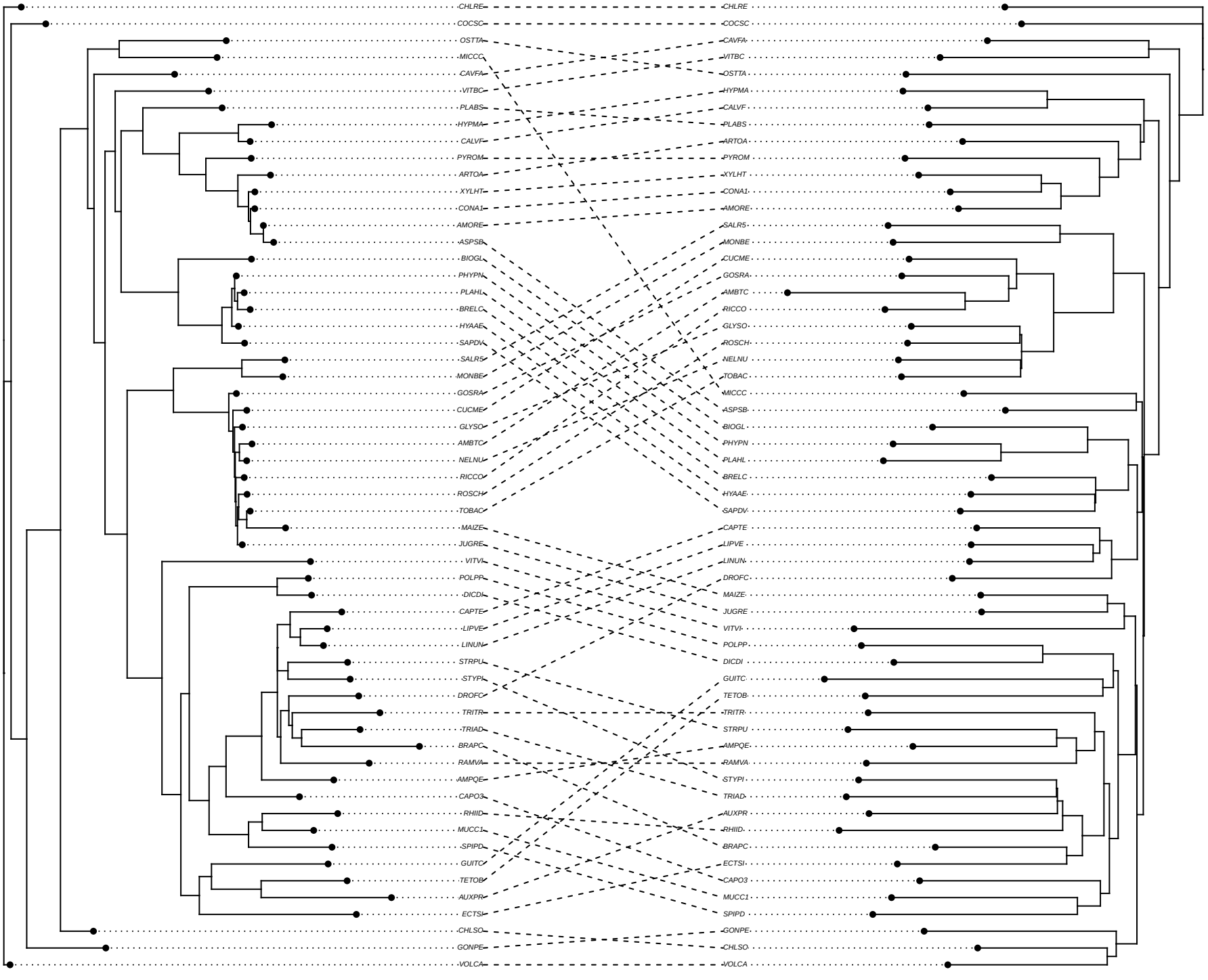
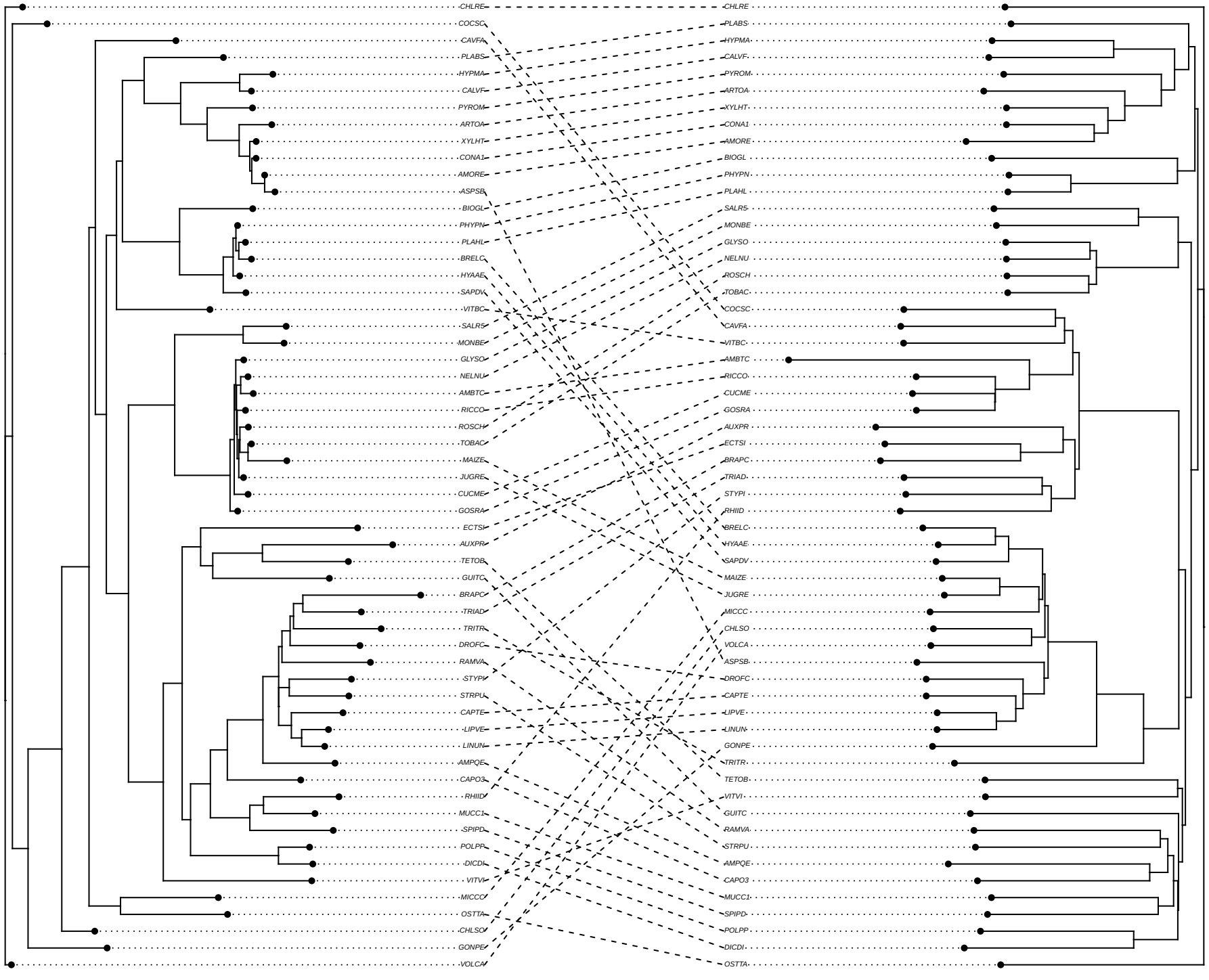

IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
EB−astral
EB−ncbi−astral
EB−ncbi
clade
Land plants
Green algae
Cryptophyta
SAR
Amoebozoa
Fungi
Holozoa
Outgroup (CHLRE)
Other


IQ−TREE vs Phylolformer
COCSC
OSTTA
MICCC
CAVFA
MONBE
SALR5
GOSRA
CUCME
GLYSO
NELNU
AMBTC
RICCO
ROSCH
TOBAC
MAIZE
JUGRE
ECTSI
AUXPR
TETOB
GUITC
RHIID
MUCC1
SPIPD
CAPO3
AMPQE
BRAPC
TRIAD
TRITR
DROFC
RAMVA
STYPI
STRPU
CAPTE
LIPVE
LINUN
DICDI
POLPP
VITVI
VITBC
SAPDV
HYAAE
BRELC
PLAHL
PHYPN
BIOGL
PLABS
CALVF
HYPMA
ASPSB
AMORE
CONA1
XYLHT
ARTOA
PYROM
CHLSO
GONPE
VOLCA
OSTTA
MONBE
SALR5
GLYSO
NELNU
TOBAC
ROSCH
GOSRA
CUCME
AMBTC
RICCO
CAVFA
COCSC
VITBC
ECTSI
BRAPC
AUXPR
RHIID
TRIAD
STYPI
MUCC1
SPIPD
CAPO3
TRITR
AMPQE
RAMVA
STRPU
GUITC
TETOB
MICCC
MAIZE
JUGRE
DROFC
LIPVE
CAPTE
LINUN
ASPSB
SAPDV
HYAAE
BRELC
CHLSO
VOLCA
GONPE
VITVI
DICDI
POLPP
PLAHL
PHYPN
BIOGL
PLABS
CALVF
HYPMA
AMORE
CONA1
XYLHT
ARTOA
PYROM

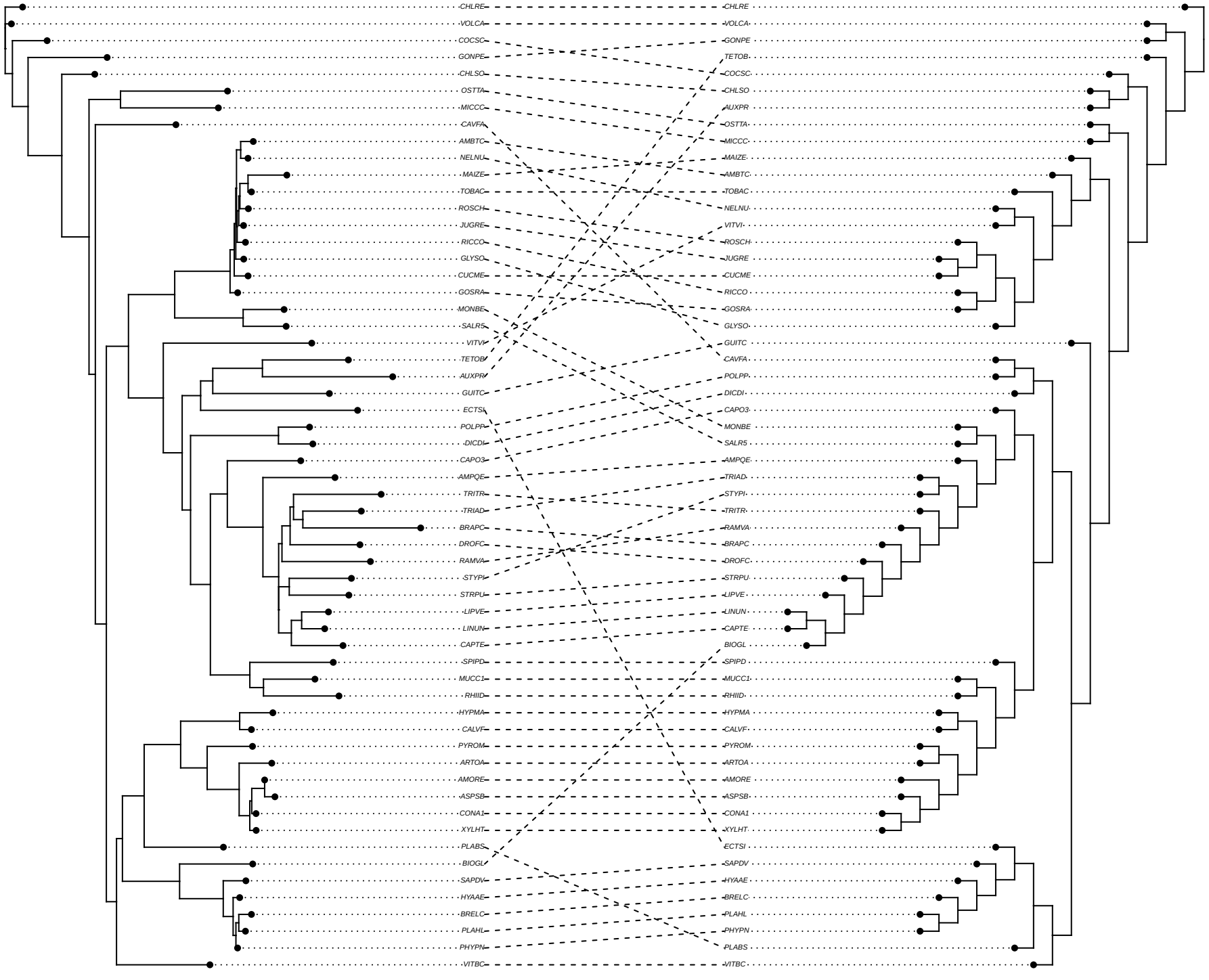
IQ-TREE vs CMAPLE



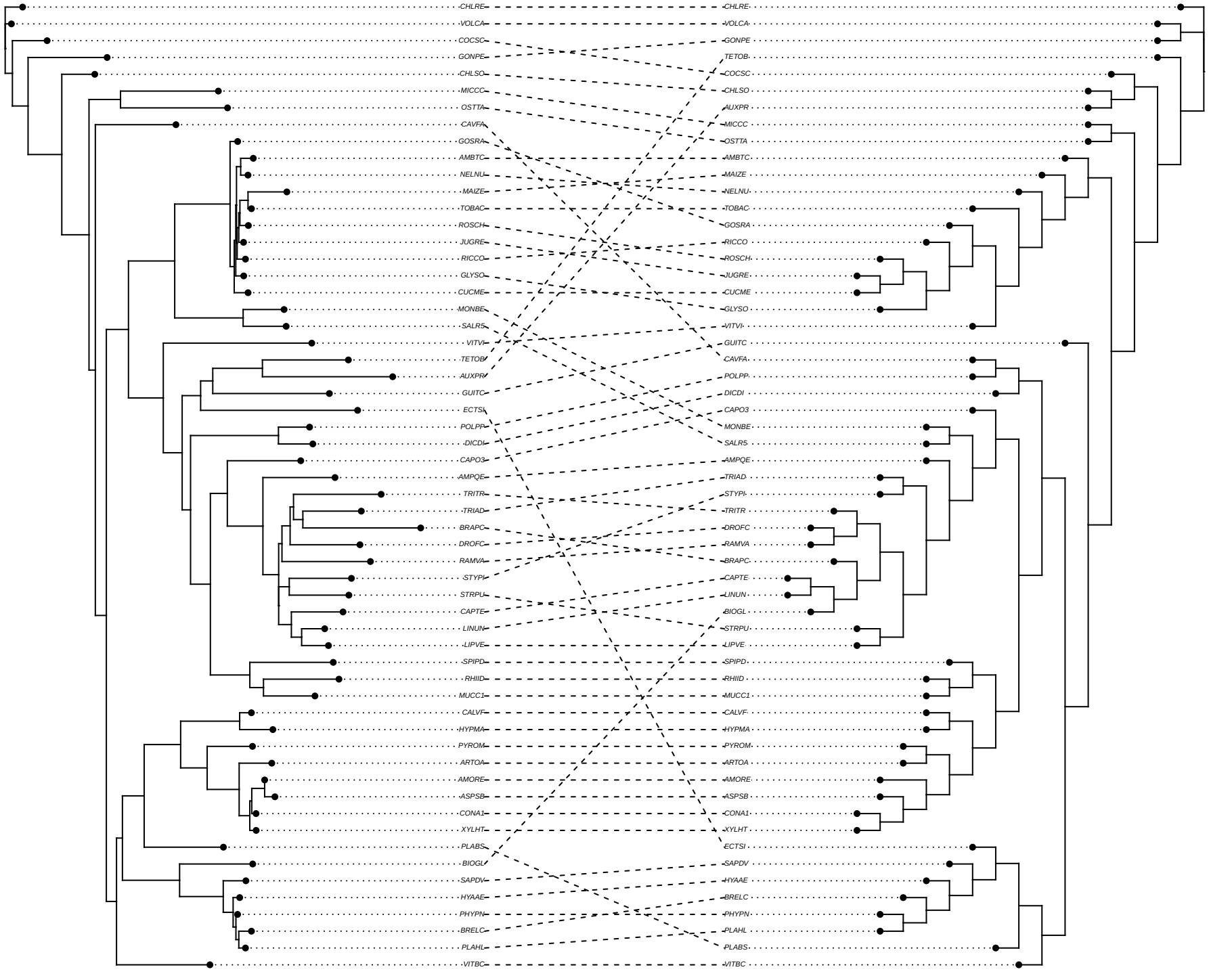

IQ−TREE vs MSA−Pattention
CHLRE
COCSC
OSTTA
MICCC
CAVFA
VITBC
PLABS
HYPMA
CALVF
PYROM
ARTOA
XYLHT
CONA1
AMORE
ASPSB
BIOGL
PHYPN
PLAHL
BRELC
HYAAE
SAPDV
SALR5
MONBE
GOSRA
CUCME
GLYSO
AMBTC
NELNU
RICCO
ROSCH
TOBAC
MAIZE
JUGRE
VITVI
POLPP
DICDI
CAPTE
LIPVE
LINUN
STRPU
STYPI
DROFC
TRITR
TRIAD
BRAPC
RAMVA
AMPQE
CAPO3
RHIID
MUCC1
SPIPD
GUITC
TETOB
AUXPR
ECTSI
CHLSO
GONPE
VOLCA
CHLRE
COCSC
CAVFA
VITBC
OSTTA
HYPMA
CALVF
PLABS
ARTOA
PYROM
XYLHT
CONA1
AMORE
SALR5
MONBE
CUCME
GOSRA
AMBTC
RICCO
GLYSO
ROSCH
NELNU
TOBAC
MICCC
ASPSB
BIOGL
PHYPN
PLAHL
BRELC
HYAAE
SAPDV
CAPTE
LIPVE
LINUN
DROFC
MAIZE
JUGRE
VITVI
POLPP
DICDI
GUITC
TETOB
TRITR
STRPU
AMPQE
RAMVA
STYPI
TRIAD
AUXPR
RHIID
BRAPC
ECTSI
CAPO3
MUCC1
SPIPD
GONPE
CHLSO
VOLCA


IQ−TREE vs MSA−T embedding
CHLRE
COCSC
CAVFA
PLABS
HYPMA
CALVF
PYROM
ARTOA
XYLHT
CONA1
AMORE
ASPSB
BIOGL
PHYPN
PLAHL
BRELC
HYAAE
SAPDV
VITBC
SALR5
MONBE
GLYSO
NELNU
AMBTC
RICCO
ROSCH
TOBAC
MAIZE
JUGRE
CUCME
GOSRA
ECTSI
AUXPR
TETOB
GUITC
BRAPC
TRIAD
TRITR
DROFC
RAMVA
STYPI
STRPU
CAPTE
LIPVE
LINUN
AMPQE
CAPO3
RHIID
MUCC1
SPIPD
POLPP
DICDI
VITVI
MICCC
OSTTA
CHLSO
GONPE
VOLCA
CHLRE
PLABS
HYPMA
CALVF
PYROM
ARTOA
XYLHT
CONA1
AMORE
BIOGL
PHYPN
PLAHL
SALR5
MONBE
GLYSO
NELNU
ROSCH
TOBAC
COCSC
CAVFA
VITBC
AMBTC
RICCO
CUCME
GOSRA
AUXPR
ECTSI
BRAPC
TRIAD
STYPI
RHIID
BRELC
HYAAE
SAPDV
MAIZE
JUGRE
MICCC
CHLSO
VOLCA
ASPSB
DROFC
CAPTE
LIPVE
LINUN
GONPE
TRITR
TETOB
VITVI
GUITC
RAMVA
STRPU
AMPQE
CAPO3
MUCC1
SPIPD
POLPP
DICDI
OSTTA

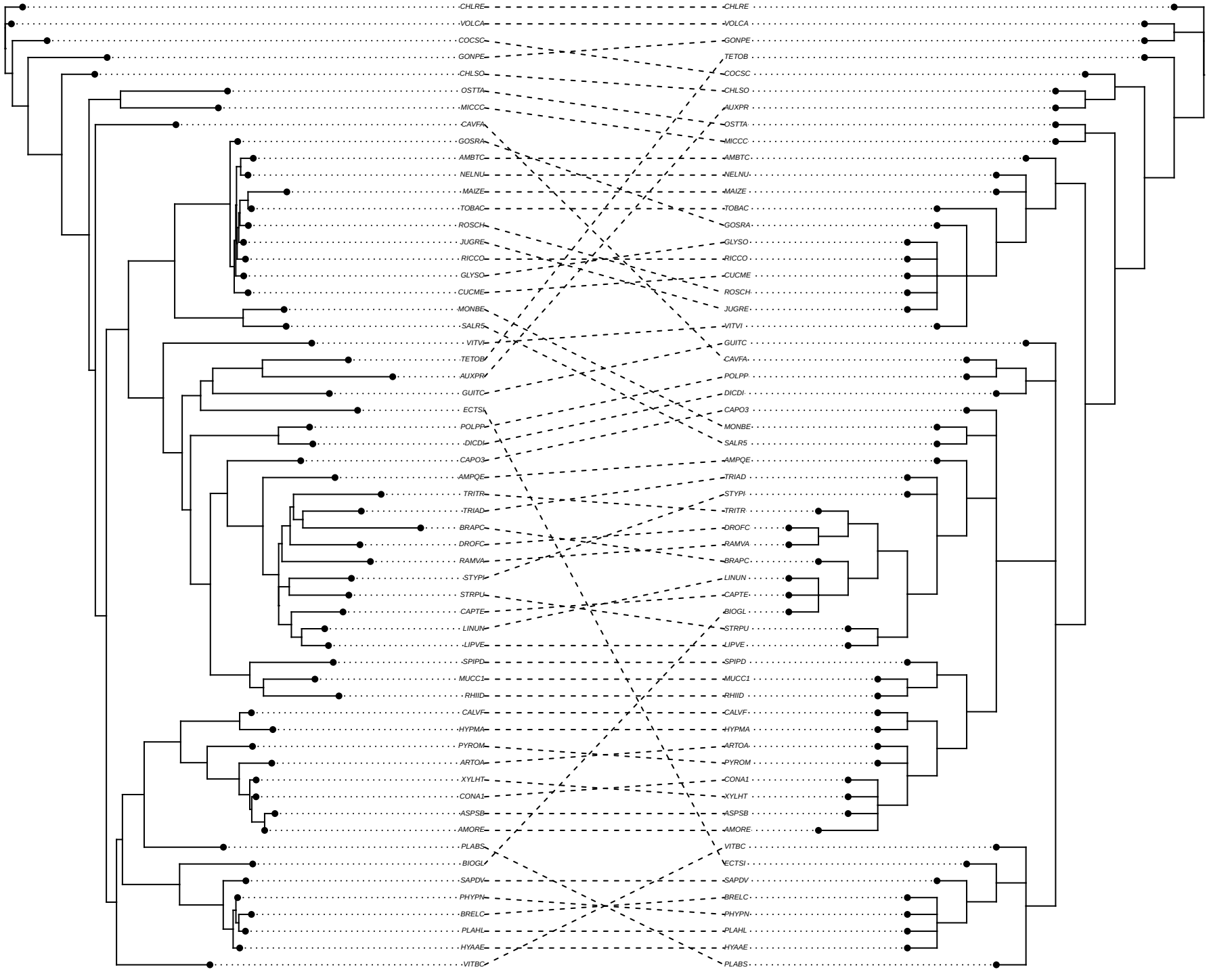
IQ−TREE vs EB−astral

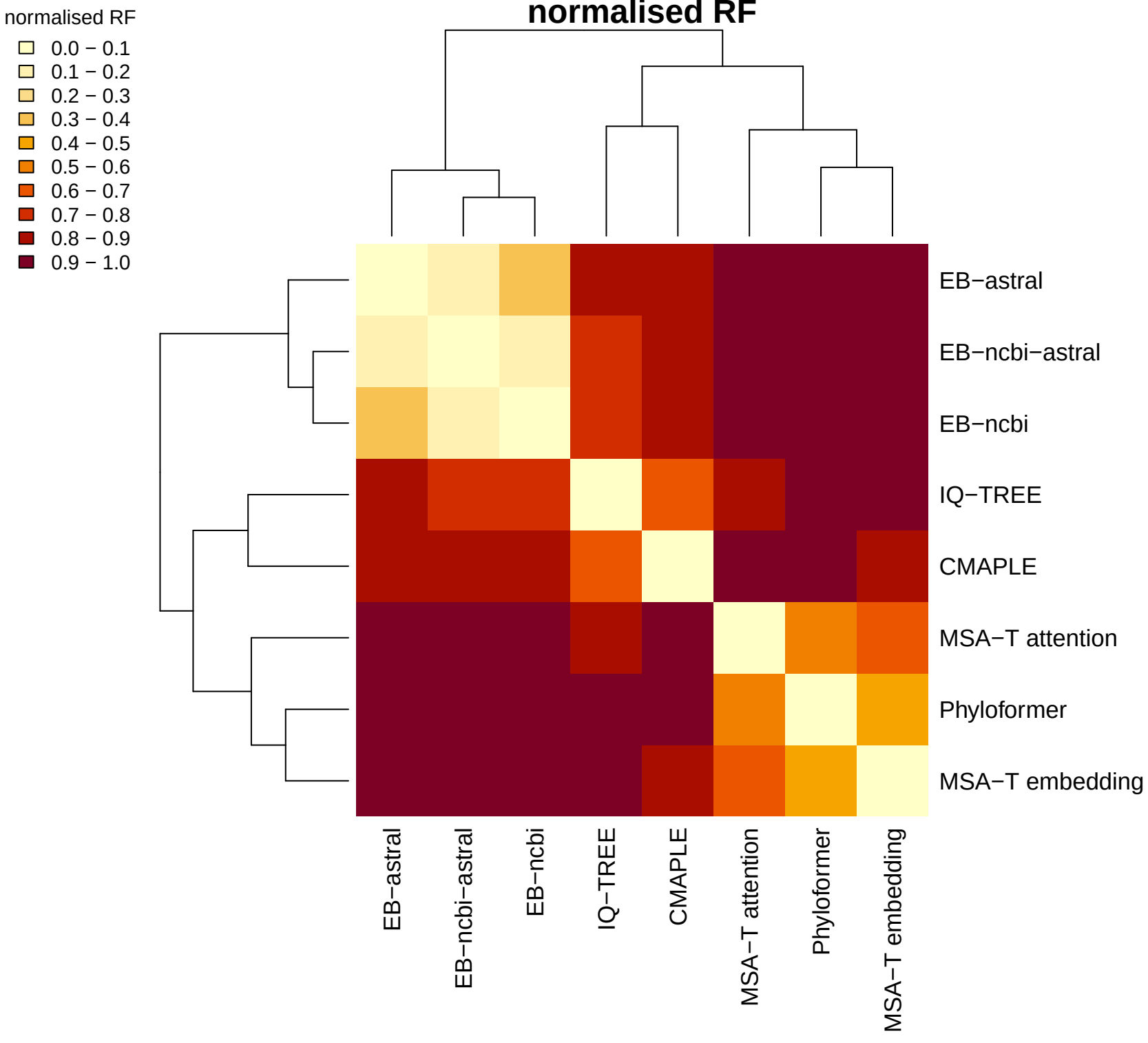


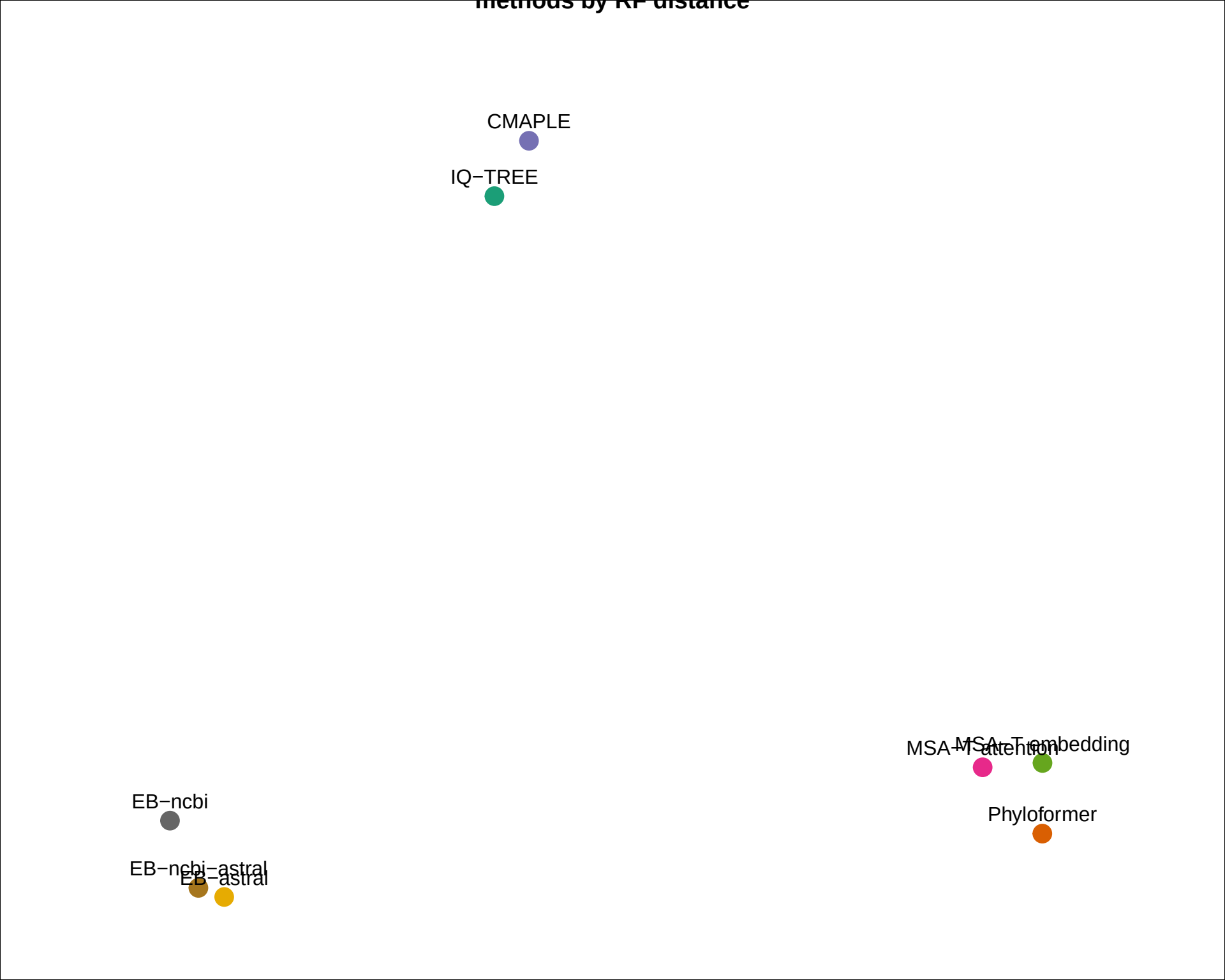
IQ−TREE vs EB−ncbi−astral

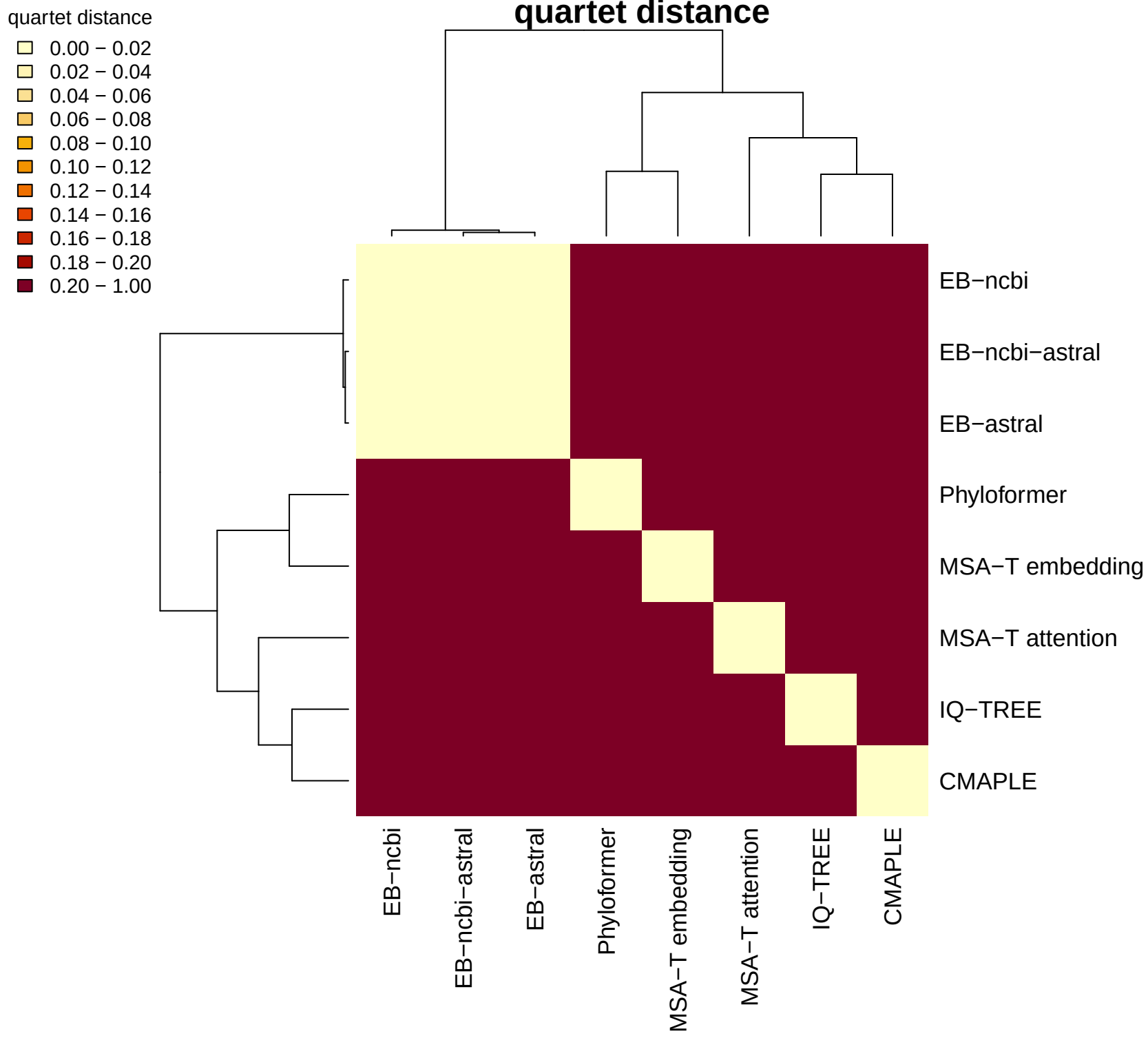


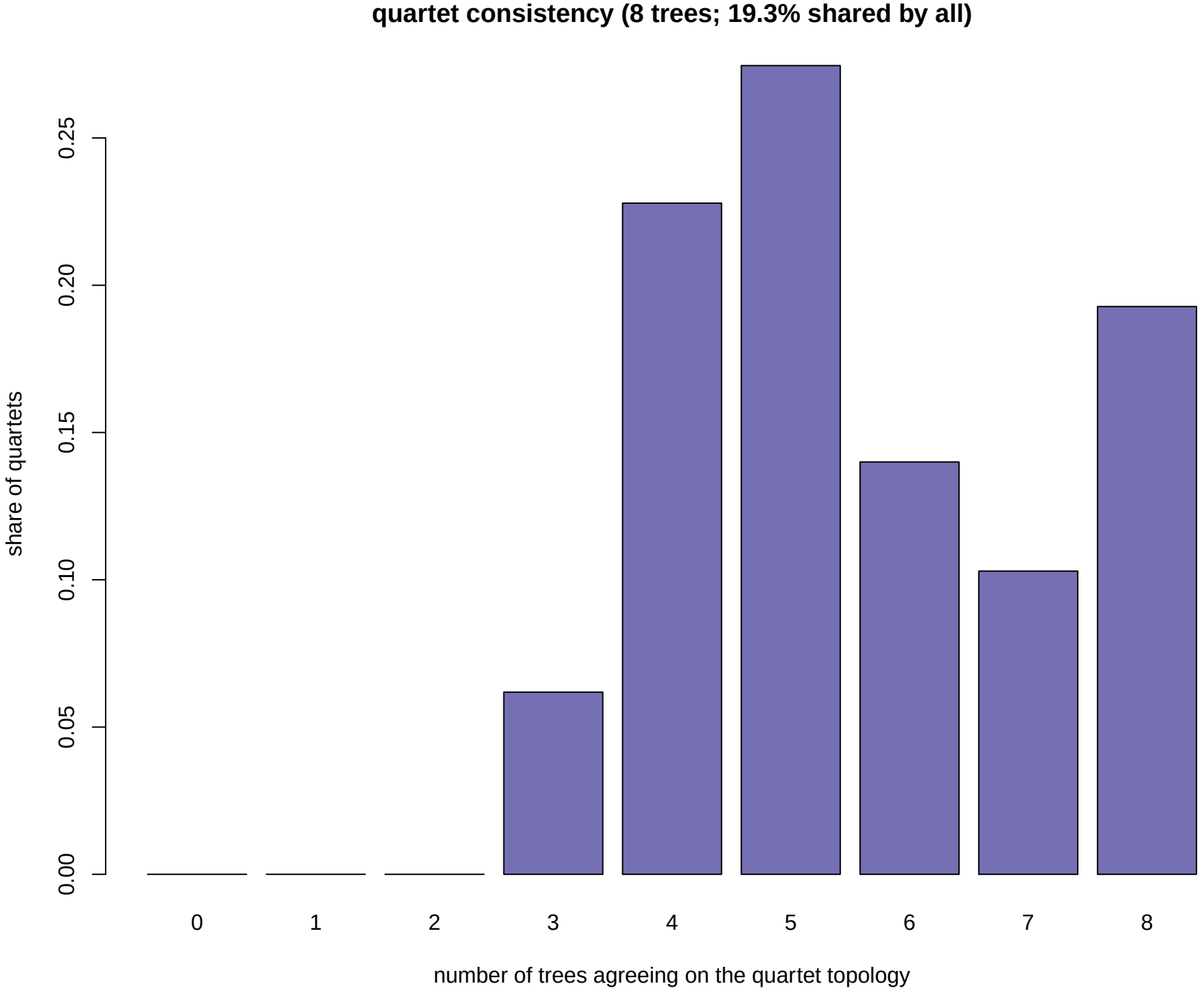
IQ-TREE vs EB-ncbi

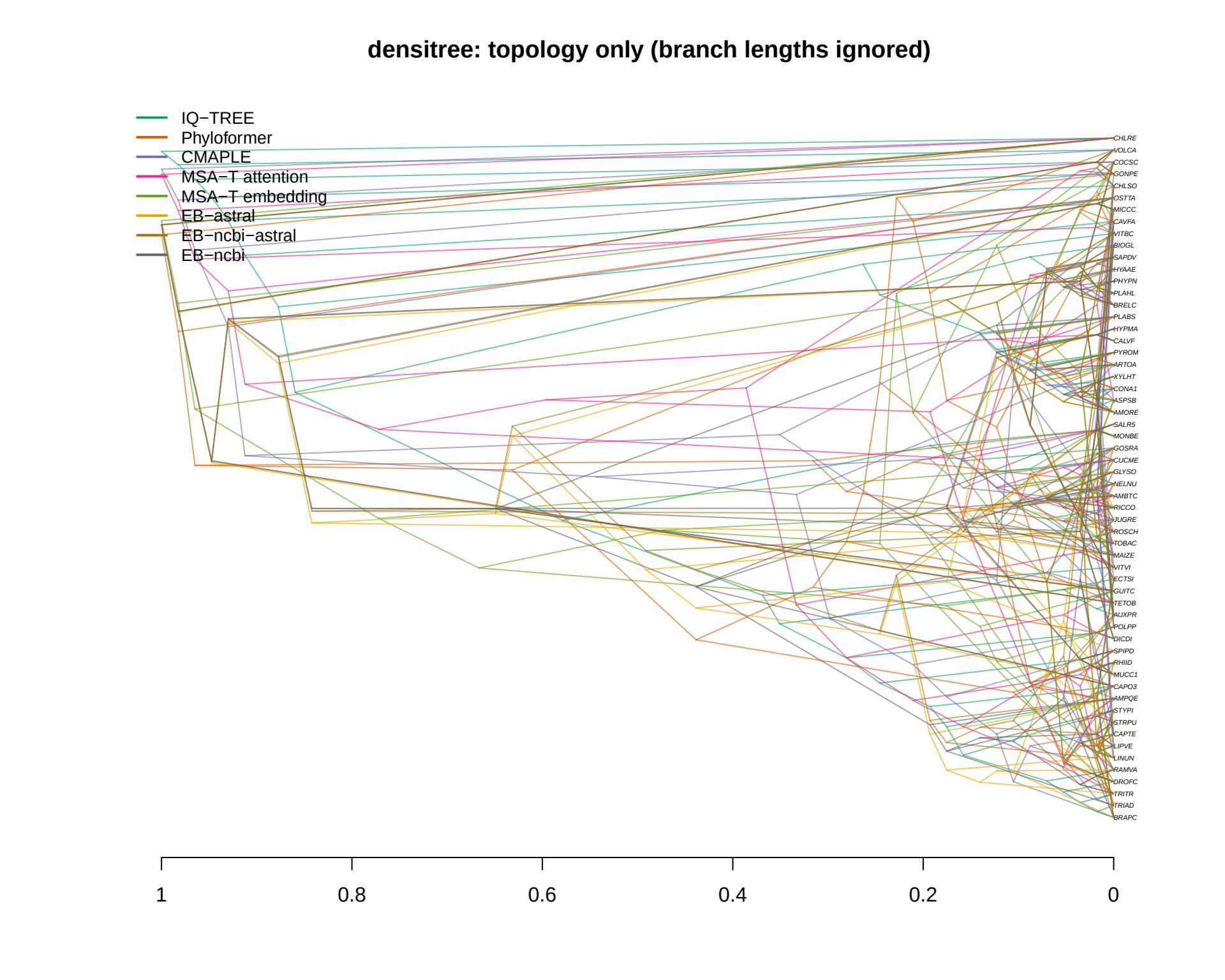


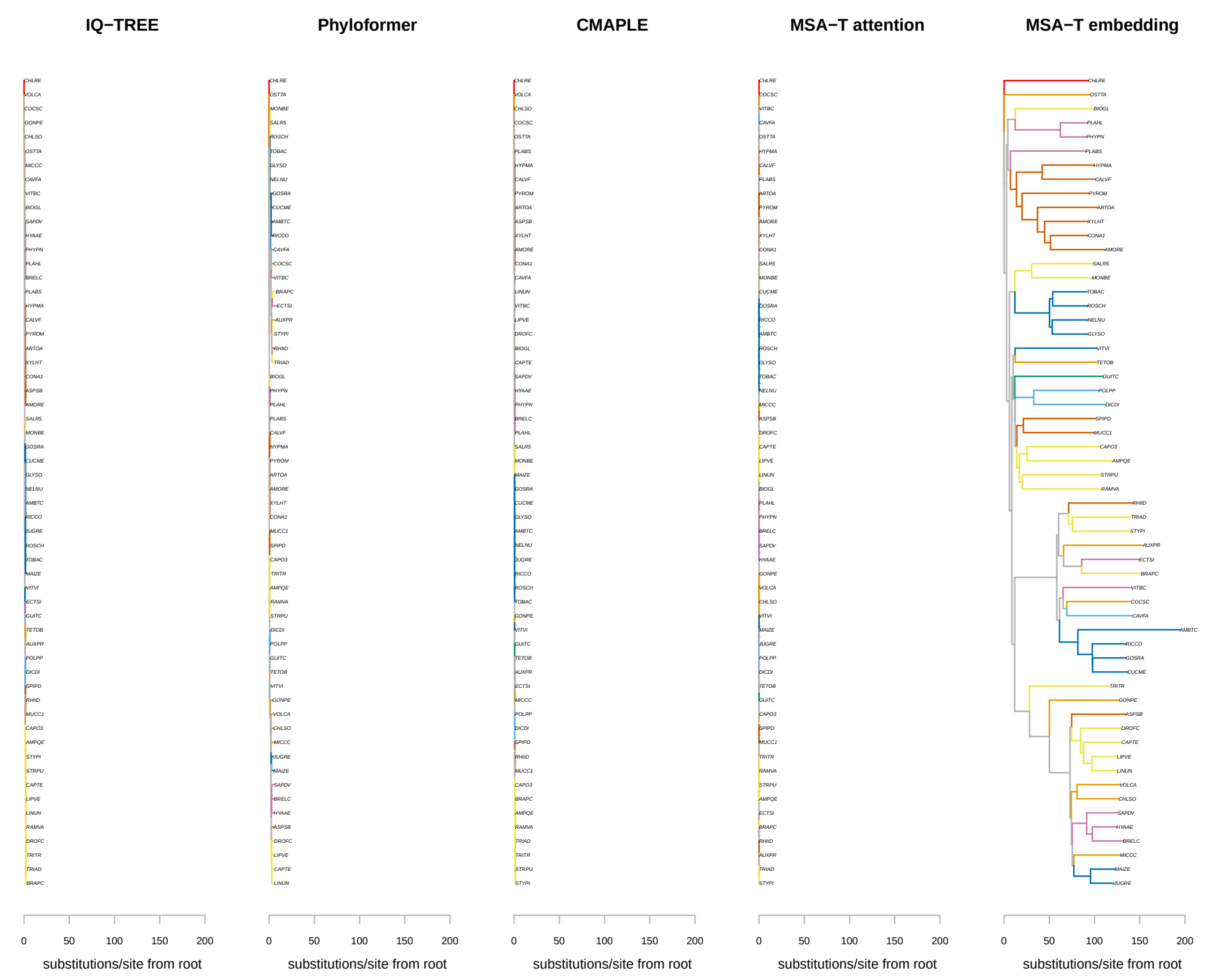

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
EB−astral
EB−ncbi−astral
EB−ncbi
IQ−TREE
CMAPLE
MSA−T attention
Phyloformer
MSA−T embedding
EB−astral
EB−ncbi−astral
EB−ncbi
IQ−TREE
CMAPLE
MSA−T attention
Phyloformer
MSA−T embedding


methods by RF distance
CMAPLE
IQ−TREE
MSA−T−embedding
MSA−T−attention
Phyloformer
EB−ncbi
EB−ncbi−astral
EB−astral

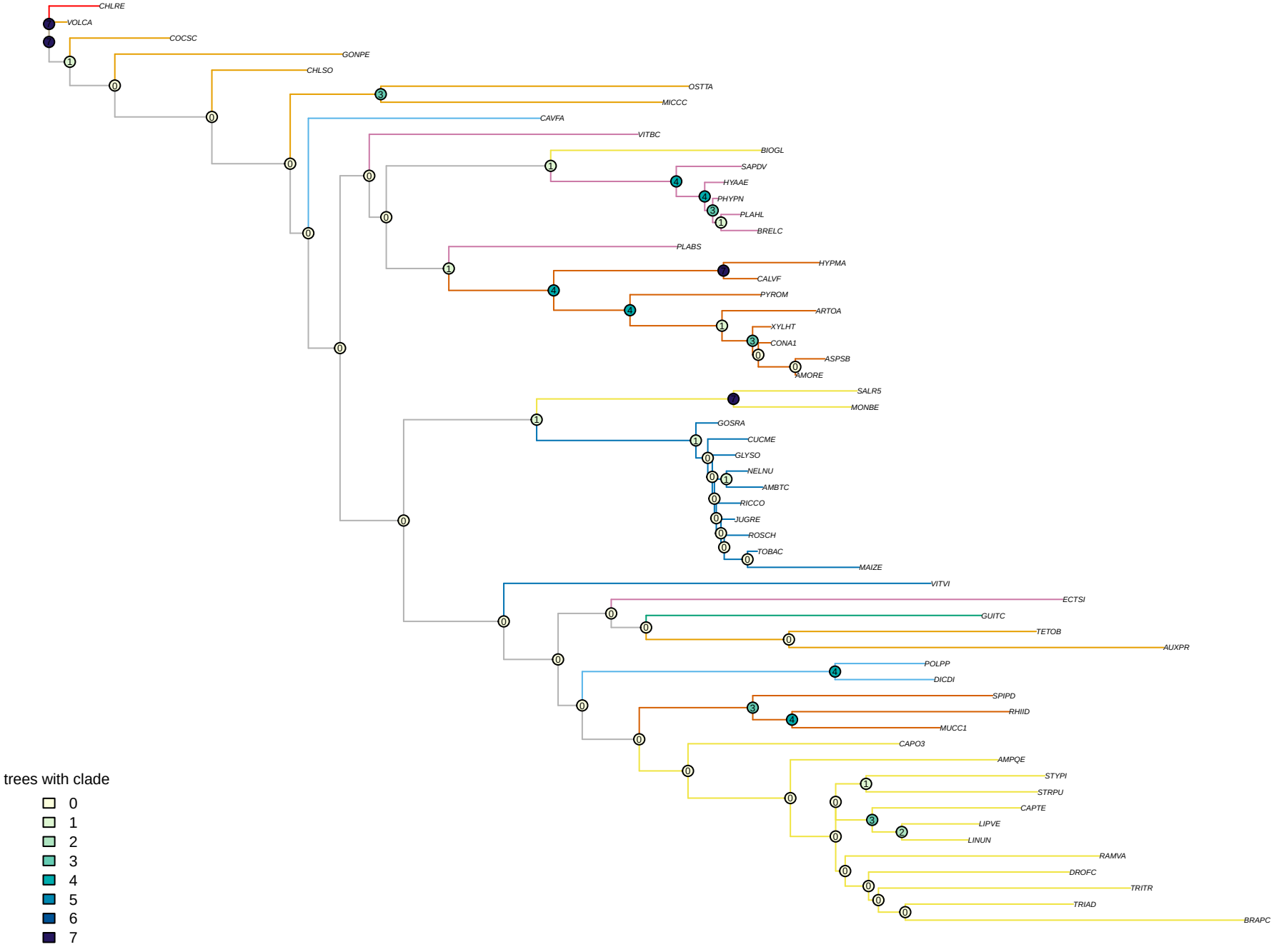

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
EB−ncbi
EB−ncbi−astral
EB−astral
Phyloformer
MSA−T embedding
MSA−T attention
IQ−TREE
CMAPLE
EB−ncbi
EB−ncbi−astral
EB−astral
Phyloformer
MSA−T embedding
MSA−T attention
IQ−TREE
CMAPLE


quartet consistency (8 trees; 19.3% shared by all)
share of quartets
0.00
0.05
0.10
0.15
0.20
0.25
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

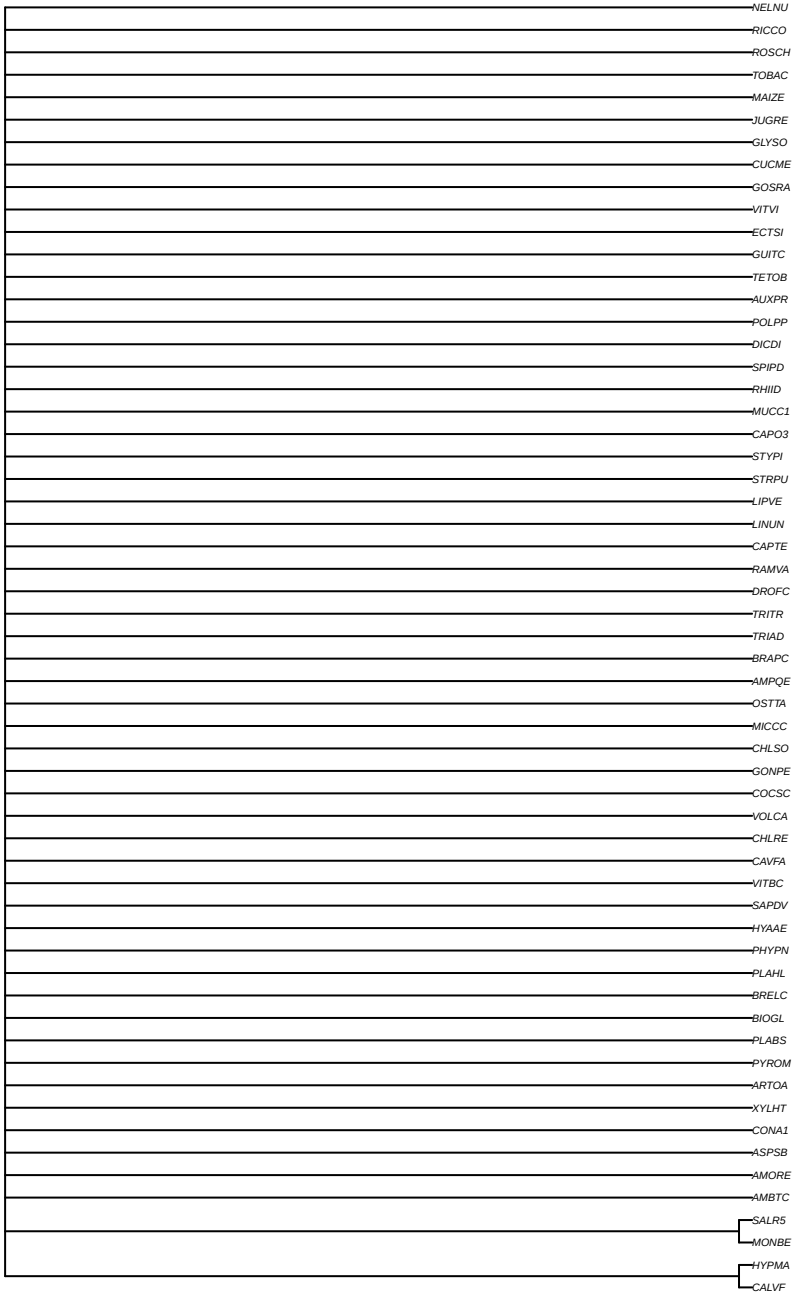

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
EB−astral
EB−ncbi−astral
EB−ncbi
CHLRE
VOLCA
COCSC
GONPE
CHLSO
OSTTA
MICCC
CAVFA
VITBC
BIOGL
SAPDV
HYAAE
PHYPN
PLAHL
BRELC
PLABS
HYPMA
CALVF
PYROM
ARTOA
XYLHT
CONA1
ASPSB
AMORE
SALR5
MONBE
GOSRA
CUCME
GLYSO
NELNU
AMBTC
RICCO
JUGRE
ROSCH
TOBAC
MAIZE
VITVI
ECTSI
GUITC
TETOB
AUXPR
POLPP
DICDI
SPIPD
RHIID
MUCC1
CAPO3
AMPQE
STYPI
STRPU
CAPTE
LIPVE
LINUN
RAMVA
DROFC
TRITR
TRIAD
BRAPC
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
CHLRE
VOLCA
COCSC
GONPE
CHLSO
OSTTA
MICCC
CAVFA
VITBC
BIOGL
SAPDV
HYAAE
PHYPN
PLAHL
BRELC
PLABS
HYPMA
CALVF
PYROM
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GUITC
TETOB
AUXPR
POLPP
DICDI
SPIPD
RHIID
MUCC1
CAPO3
AMPQE
STYPI
STRPU
CAPTE
LIPVE
LINUN
RAMVA
DROFC
TRITR
TRIAD
BRAPC
0
50
100
150
200
substitutions/site from root

IQ−TREE clades: number of the other 7 trees that contain each clade



strict



majority rule

